CSC311: Introduction to Machine Learning

Project Report

Optional Creative Project Title

Submitted by

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1 Executive Summary

We propose to classify a sport as soccer, baseball, or basketball using survey responses about its accessibility, playing characteristics, and cultural associations. The dataset contains 870 responses from 290 respondents, with 290 examples of each sport. We will compare regularized softmax regression, random forests, and Naive Bayes to investigate how linear relationships, nonlinear interactions, and textual evidence contribute to classification. We hypothesize that softmax regression will provide a strong balance between predictive performance and interpretability when trained on encoded survey responses and text features. We will select the final model using validation results grouped by respondent and assess it using accuracy and macro-F1 which is used for multi-class and multi-label classification models. A projected test accuracy will be reported after completing these experiments.

2 Data Exploration

Dataset Overview and Splitting

- **Dataset Composition**: The dataset contains 870 survey responses collected from 290 distinct students, with each respondent evaluating all three sports (Soccer, Baseball, and Basketball).
- **Class Balance**: The target labels are perfectly balanced (290 observations per sport overall; exactly 203 per sport in the 70% training split). No class rebalancing or oversampling is needed.
- **Respondent Isolation (`id`)**: Each respondent's unique identifier is excluded from the feature matrix X and used exclusively for grouped partitioning. Keeping all responses from the same participant within the same split prevents the model from memorizing respondent-specific vocabulary, bias, or stylistic quirks.
- **Data Split**: The 290 respondent IDs are partitioned into the following splits by repondent to remove leakage from memorizing specific respondent answers:
  - **Training split (70%)**: 203 respondents (609 observations) used for exploratory analysis, distribution inspection, and parameter estimation.
  - **Validation split (15%)**: 43 respondents (129 observations) reserved strictly for hyperparameter tuning.
  - **Internal test split (15%)**: 44 respondents (132 observations) untouched until final model evaluation.

  All statistical parameters (imputation medians, standard scaling means/stds, and text vocabularies) are learned strictly on the 70% training split.

Feature Preprocessing Pipeline

Table 1 summarizes the 12 survey features, identifier handling, target labels, and machine learning rationales.

Table 1: Comprehensive breakdown of all 14 survey features, preprocessing pipeline, and machine learning rationales.

Survey Feature	Raw Type	Preprocessing Pipeline	Final Representation	Machine Learning Rationale & Advantage
Respondent ID (`id`)	Categorical ID	Excluded from features; used for grouped partition	Grouping Index (not in X)	Prevents style memorization; ensures zero leakage across folds.
Ease of Playing (`difficulty_rating`)	Numeric (0–10)	Mean imputation ($\mu = 5.48$); Z-score standardization	Standardized continuous	Eliminates scale disparity; stabilizes gradient descent convergence.
Strategy Importance (`strategy_importance`)	5-point Likert	Monotonic mapping (1–5); neutral imputation (3)	Ordinal scalar $\in \{1..5\}$	Preserves ordered agreement degrees without inflating dimensionality.
Injury Risk Comfort (`injury_risk_comfort`)	5-point Likert	Monotonic mapping (1–5); neutral imputation (3)	Ordinal scalar $\in \{1..5\}$	Baseball sits about one point below soccer and basketball on the ordered scale.
Individual Brilliance (`individual_brilliance`)	5-point Likert	Monotonic mapping (1–5); neutral imputation (3)	Ordinal scalar $\in \{1..5\}$	Soccer sits lower than baseball and basketball; those two are close.
Playing Locations (`play_location`)	Multi-select CSV	Lowercase; split by delimiter; binary multi-hot expansion	Binary vector (7 dims)	Accommodates concurrent, non-exclusive venues (e.g., school and park).
Toronto Seasons (`play_season`)	Multi-select CSV	Lowercase; split seasons; binary multi-hot expansion	Binary vector (4 dims)	Models multi-season playability (e.g., summer/fall outdoor vs winter).
Music Genres (`associated_music`)	Multi-select CSV	Lowercase; split genres; binary multi-hot expansion	Binary vector (10 dims)	Captures non-exclusive cultural affinities across musical genres.
Foods and Drinks (`associated_foods_drinks`)	Free-form text	Lowercase; strip punctuation; TF-IDF downweighting; missing flag	Sparse TF-IDF + missing flag	**TF-IDF**: Penalizes generic filler ("the", "with"); boosts terms like "popcorn".
Associated Colour (`associated_color`)	Free-form text	Lowercase; clean punctuation; TF-IDF representation; missing flag	TF-IDF vector + missing flag	Unifies casing ("Orange" vs "orange"); isolates discriminative colors.
Associated Song (`associated_song`)	Free-form text	Lowercase; clean punctuation; TF-IDF representation; missing flag	Sparse TF-IDF + missing flag	Blank songs are less common for soccer than for baseball or basketball.
Emotional Response (`emotional_response`)	Free-form text	Lowercase; strip punctuation; TF-IDF vectorization; missing flag	Sparse TF-IDF + missing flag	Differentiates excitement vs boredom while filtering non-informative words.
Willingness to Pay (`acceptable_price`)	Unstructured text	Regex strip "$", commas; convert "thousand"; median fill; $\log(1+x)$	Log continuous scalar	Compresses a right tail from $0 to $10,000 (training skewness about +7 to about −1.2).
Target Sport (`label`)	Class string	Map Soccer $\to 0$, Baseball $\to 1$, Basketball $\to 2$	Integer label $y \in \{0, 1, 2\}$	Standard multi-class target; perfectly balanced (203 per class in train split).

Key Preprocessing Strategies and Rationale

- **TF-IDF vs. Raw Word Counts**: In free-text survey responses, uninformative words such as "the", "a", and "with" appear frequently across all three sports. A simple term-frequency counter would incorrectly assign these generic words the highest weights. TF-IDF applies an inverse-document-frequency penalty, heavily unweighting common words while placing more importance on discriminative, sport-specific terms (e.g., "pitcher", "popcorn", "court").
- **Handling Extreme Outliers with** $\log(1 + x)$: Parsed ticket prices on the training split run from $0 to $10,000 and are strongly right-skewed (skewness about +7). Applying $\log(1 + \text{price})$ compresses that tail (skewness about −1.2). The transformed values are still mildly left-skewed, but extreme tickets no longer dominate a linear model.
- **Informative Missingness Indicators**: On the training split, most fields are blank in under 2% of rows. Foods and drinks are blank in about 3%. The song field is blank in about 10% of baseball and basketball responses and about 3% of soccer responses, so the omission itself is related to the label. A binary indicator (`is_missing`) is appended to each text feature instead of dropping those rows.

Feature–Label Patterns

The choices in Table 1 come from the training split. Difficulty is already a 0–10 number spread across the full scale, so the training mean (5.48) fills the few blanks and a z-score puts it on a comparable scale to the other inputs. The Likert answers are ordered, and the class shifts below move in one direction along that order, so we keep a single 1–5 code and fill blanks with Neutral rather than one-hot encoding. Multi-select answers can contain several options at once, so each observed option becomes its own binary column: 7 locations, 4 seasons, and 10 music genres. Free-text answers are lowercased and stripped of punctuation, then represented with TF-IDF plus a missingness flag, because words such as "the" and "with" would otherwise receive the largest weights. Prices are parsed out of free text before the log transform above. About 8% of training prices do not parse, at similar rates for all three sports, and those entries take the training median.

Table 2 reports training-split means by sport. Standard deviations within a sport are about 2.0 for difficulty, about 0.8–1.1 for each Likert item, and about 1.6 for log price.

Ease of playing and injury-risk comfort separate baseball from the other two sports by about one within-class standard deviation. Individual brilliance separates soccer, which sits lower,

Table 2: Training-split means by sport. Log price is computed only on responses that parsed as a number.

Feature	Baseball	Basketball	Soccer
Ease of playing (0–10)	4.18	6.05	6.22
Strategy importance (1–5)	3.45	3.92	3.88
Injury-risk comfort (1–5)	2.85	3.81	3.77
Individual brilliance (1–5)	3.74	3.61	2.93
Log ticket price	3.85	4.71	4.84

from baseball and basketball, which are close to each other. Strategy and log price move in the same direction, but the gaps are smaller than the spread inside each sport, so they are weaker signals. The Likert answers shift in one direction rather than forming a U-shape. For injury-risk comfort, baseball answers concentrate on Disagree and Neutral, while soccer and basketball concentrate on Agree. Correlations among these five features are modest. The largest is 0.31, between strategy and injury-risk comfort, so the ordered items are not copies of one another.

The multi-select items separate the sports more sharply than the Likert items. On the training split, winter is selected in 61% of basketball responses, 9% of soccer responses, and 5% of baseball responses. An indoor location is selected in 95% of basketball responses and 27% of baseball responses. Latin music is selected in 39% of soccer responses and in under 4% of the other two sports. Country music is selected in 28% of baseball responses and in under 5% of the others. Hip-hop or rap is selected in 83% of basketball responses, 41% of soccer responses, and 25% of baseball responses. “No particular type” of music is selected in 32% of baseball responses and 7% of basketball responses. These are large differences in binary rates, so a linear classifier can use the multi-hot columns directly. Winter is uncommon overall and still highly predictive, because almost all of those selections belong to basketball.

Song nonresponse follows the label as well: about 10% of baseball and basketball training responses leave the song blank, compared with about 3% of soccer responses. That is the reason to keep a missingness indicator. The other text fields are too complete for missingness alone to carry much signal.

These patterns are why Section 3 compares regularized softmax regression, a random forest, and Multinomial Naive Bayes. The dominant pattern is a shift in a mean or a selection rate, which a linear softmax model can represent, and L2 regularization matters once TF-IDF turns the text into a wide sparse matrix relative to 609 training rows. A random forest is still worth fitting because the strongest cues can combine. Basketball, for example, is associated with both winter and indoor locations, and a tree can split on that pair even though each margin is already informative. We did not find a Likert item whose middle values belong to one sport and whose extremes belong to another, so the forest is included for interactions rather than for non-monotonic main effects. Naive Bayes is the text baseline for the remaining case: no single token is expected to separate the classes, but many weak

word-level cues can add up. Because several structured features already separate the sports, each family will be compared on structured features, text features, and both together.

3 Methodology

Data splitting and validation

We reserve 15% of respondent IDs as an internal test set, 15% as a validation set, and 70% for model training. On the remaining respondents, we will use five-fold grouped cross-validation for hyperparameter tuning. All three responses from each respondent will remain together throughout this process. Imputation values, scaling parameters, categorical vocabularies, and text vocabularies will be learned separately from each fold's training data. The internal test set will remain unused during feature selection and hyperparameter tuning. Model 1: Regularized softmax regression Softmax regression will predict the probabilities of the three sports from a combination of structured features and TF-IDF text features. It is appropriate as a relatively simple model for a dataset with many potentially sparse text features. L2 regularization will help control overfitting. We will tune the regularization parameter, compare unigrams with unigrams plus bigrams, and vary the minimum document frequency used to retain text features. We will inspect model coefficients to identify which responses contribute most strongly to each prediction. Model 2: Random forest A random forest will model nonlinear relationships and interactions between responses. For example, the relationship between a season response and a sport may depend on the locations selected by the respondent. We will first evaluate the forest using structured features, then test whether adding a restricted text vocabulary improves validation performance. We will tune maximum tree depth, minimum samples per leaf, and the number of features considered at each split. Candidate settings could include 100 and 300 trees, maximum depths of 5, 10, and unrestricted depth, and minimum leaf sizes of 1, 3, and 5. Model 3: Naive Bayes Multinomial Naive Bayes will provide a probabilistic text baseline using non-negative word-count or TF-IDF features from the free-text questions. It is computationally inexpensive and can combine many weak word-level signals. We will tune the smoothing parameter over values such as 0.1, 0.5, 1, and 2, and compare word counts with TF-IDF. Its conditional-independence assumption may limit performance when words or survey responses are strongly related.

Fair comparison and feature experiments

All model families will use the same respondent-based folds and receive hyperparameter tuning before comparison. We will compare structured-only, text-only, and combined representations where appropriate to determine how much each feature group contributes. The supplied kNN implementation can serve as an additional reference baseline. We will also report a constant-class baseline, which achieves 33.3% accuracy on this balanced dataset. Evaluation We will report accuracy, macro-F1, per-class precision and recall, and confusion matrices. Accuracy measures overall correctness, while macro-F1 gives each sport equal importance and helps reveal uneven performance across classes. We will summarize the mean and standard deviation of cross-validation scores. Our primary selection metric will

be mean validation accuracy, with macro-F1 and consistency across folds informing close comparisons.

4 Results

For the proposal, use planned analysis rather than invented scores:

We will summarize the tuned models' cross-validation performance in the following table:

TABLE

We will analyze confusion matrices to determine which sports are most frequently confused. We will also examine whether errors are associated with missing responses, uncommon song titles, ambiguous text, or similar structured answers.

We hypothesize that softmax regression will perform well because regularization can help it handle sparse text features. Random forests may perform better if interactions among structured responses are especially informative. Naive Bayes will establish how effectively word-level evidence alone distinguishes the classes.

After selecting the model and preprocessing configuration, we will evaluate the selected pipeline once on the internal test set. We will use this result, supported by cross-validation stability, to justify a single projected accuracy for the unseen course test set. No defensible numerical estimate is available yet.

5 Contributions and Learning

Bianca -

Sylvia -

Chris -

6 References

Use references whenever appropriate (e.g., when you consult textbooks, research papers, online tutorials, or external code). Cite in the text using `\cite{}`. For example, let's cite the two references provided in the `.bib` file [**?**, **?**].